

MASTER IN DATA SCIENCE AND ADVANCED ANALYTICS
DEEP LEARNING

SKIN LESSIONS

A STUDY ON DERMATOSCOPIC IMAGES

[GITHIB REPOSITORY]

[DATASET SOURCE]

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1. INTRODUCTION

The integration of Deep Learning into medical practice has grown substantially since 2012, driven by its ability to scale with large datasets, leverage specialized hardware, and continuously improve with additional data, often surpassing classical machine learning approaches.

A key strength of Deep Learning lies in its capacity to process multimodal inputs, such as medical images, electronic health records and genomic data, enabling a more comprehensive representation of a patient's condition. In the domain of medical imaging, in particular, Convolutional Neural Networks (CNNs) have proven especially effective, exploiting the spatial structure of images to detect subtle visual patterns that may escape the human eye [\[1.\]](#).

Skin lesion analysis fits into this landscape, as dermatological diagnosis has always depended on visual inspection, making it a well-suited problem for image-based deep learning. The motivation behind the present work is to explore how deep learning can be applied to dermoscopic image classification of common pigmented skin lesions.

2. DATASET

This work uses the HAM10000 (*Human Against Machine with 10000 training images*) dataset [\[2.\]](#), a large collection of dermoscopic images assembled from different populations and with various dermatoscope types. It comprises over 11,000 images covering 7 diagnostic categories of pigmented skin lesions ([Table 1](#)).

3. DATA UNDERSTANDING AND PREPROCESSING

3.1. EXPLORATORY DATA ANALYSIS (EDA)

The first step was to do an exploratory data analysis to understand the dataset before making any modelling decisions. The class distribution showed a strong imbalance, with nevus (nv) representing 66% of samples while minority classes like dermatofibroma and vascular lesions account for less than 2% each, as seen in [Figure 1](#). Every image has a fixed resolution of 600 pixels in width and 450 pixels in height.

After an analysis on the demographic patterns, we observed that certain lesion types were associated with specific age groups and that the sex distribution is roughly balanced ([Figure 2](#)). The body location analysis showed that most lesions concentrate on the back and lower extremities, with notable variation across diagnoses ([Figure 3](#)). We inspected sample images from each class to assess quality and the presence of artifacts and examined the lesion ID structure to identify cases where multiple images

correspond to the same lesion to prevent data leakage in train/test splits. We also reviewed the average image per class diagnosis, that showed that visual features like pigmentation intensity and blood concentration are indicators for differentiate across classes ([Figure 4](#)); and reviewed the confirmation methods (histopathology, follow-up, consensus, confocal), noting varying label reliability across classes.

Finally, we used UMAP projections on pretrained model embeddings to visualize the separability of inter classes and ensured that duplicated lesion cases were properly handled to avoid information leakage.

3.2. DATA PREPROCESSING

Preprocessing aims to address data quality issues, prevent information leakage, and thus build the foundation upon the models will be created. Our pipeline proceeded in principled stages, each tackling specific challenges of dermatoscopic image classification.

3.2.1. Data Integrity and Stratified Data Split

The workflow began with image-level deduplication to remove exact pixel-level duplicates, which can artificially inflate model performance. More critically, when splitting the dataset, we aimed to prevent information leakage by verifying lesion IDs across all samples and ensuring no overlapping identifiers between splits. Multiple images from the same lesion were kept together in a single split, preserving the independence assumption essential for reliable evaluation. We applied an 80-10-10 stratified partition, guaranteeing balanced class representation across train/validation/test sets.

Age information contained 263 missing values, which were handled via iterative imputation (`IterativeImputer`). This strategy outperformed simpler alternatives ([Figure 5](#)) by leveraging correlations between demographic and lesion-type features, thereby retaining data richness while maintaining statistical validity.

3.2.2. Image Cleaning and Standardization

Dermatoscopic images frequently contain occlusive artifacts, such as hairs, that are manifested as high-contrast linear structures overlaid on lesion. We applied the `DullRazor` algorithm [\[3.\]](#), a method that identifies and removes hair through smooth interpolation of surrounding skin ([Figure 6](#)). Generic enhancement techniques such as CLAHE and background removal were also considered, but deemed unnecessary, as the subsequent augmentation strategy (jittering, random erasing) already addresses the brightness, contrast variations that these techniques target. UMAP projections on pretrained embeddings were used to visualize inter-class separability and semantic outliers, though these were retained in the dataset to preserve population diversity.

All images in HAM10000 share uniform native resolution (600x450 pixels). Our chosen approach was to standardize inputs to 224x224 pixels, a size small enough to

accelerate training but big enough to preserve morphological details. Critically, this resolution matches ImageNet pretraining standards, allowing direct transfer of feature representations from the pretrained models used. Among the tested resizing approaches, center-crop was selected because it preserved natural lesion shapes and aspect ratios, and kept diagnostic features intact without introducing artificial padding or distortion like the alternatives, as shown in [Figure 7](#).

3.3. DATA AUGMENTATION

3.3.1. Performance comparison between online and offline augmentation

Prior to implementing the final data augmentation pipeline, we conducted a comparative study to evaluate the viability of offline augmentation (pre-generating and saving synthetic images to disk) versus online augmentation (applying real-time transformations per batch during training). Given the time constraints for the project, optimizing the time required to run and train the models became a priority. Even though generating the augmented images beforehand required an initial upfront time of investment, it was essential for accelerating the overall experimentation process. By eliminating the computational bottleneck of applying transformations on the fly during each batch, the offline method reduced epoch training time by approximately 42% and nearly doubled the processing speed per step, as shown in [Table 2](#).

Ultimately, offline augmentation was selected as the augmentation strategy due to: the faster training time that enables rapid and iterative experimentation; the final validation accuracy that remained highly competitive with the online approach; the training instability observed that was resolved upon transitioning to larger better capacity models; and offline augmentation was the only structure capable of executing customized transformations aggressively targeted per class, essential for addressing our class imbalance.

3.3.2. Final offline augmentation strategy

Instead of balancing all classes to the exact size of the majority class, which would risk overfitting in the rarest categories like dermatofibroma (df), we employed a square root balancing method. First, we established a target size by undersampling the majority class (nevus) to approximately 70% of its original volume. For each minority class, an augmentation multiplier was then calculated as the rounded square root of the ratio between this majority target size and the minority class count. This approach naturally smoothers the class distribution, boosting the representation of minority classes, and preserving the essential distribution of the real world population. The final distribution is shown on [Figure 8](#).

To generate synthetic samples, we designed a transformation strategy per class. We assigned specific levels of aggressiveness based on the severity of a class's

underrepresentation, ensuring that classes requiring higher multipliers were subjected to more complex spatial and color distortions to maximize feature variance:

- Strategy 1 (light): applied to moderately imbalanced classes (such as *bkl*). It focuses strictly on geometric transformations, specifically random horizontal and vertical flips, alongside rotations. This tier preserves the original morphological structures and color distributions of the lesions.
- Strategy 2 (moderate): utilized for classes with higher scaling needs (such as *akiec*). In addition to geometric transformations, this tier introduces mild color jittering (adjusting brightness and contrast) to simulate different real world dermatoscopic lighting conditions.
- Strategy 3 (aggressive): reserved for the most extreme minority classes (such as *df*). This strategy combines the aforementioned augmentations with heavy spatial distortions, including elastic transforms, Gaussian blur, and random perspective shifts. It also incorporates random erasing to simulate artifacts such as hair or immersion of fluid occlusion.

Through this augmentation strategy, the dataset was artificially expanded and rebalanced, establishing a much more robust base for deep learning models to generalize effectively across all diagnostic categories.

4. MODEL DEVELOPMENT

All models were developed and trained on a single workstation running Python 3.11, a requirement imposed by NVIDIA GPU driver compatibility with TensorFlow. This constraint, combined with strict library dependencies, meant that all experiments were run sequentially on a single machine, limiting the breadth of simultaneous experimentation.

4.1. EVALUATION STRATEGIES AND METRICS

Given the severe class imbalance identified in Section 2, raw accuracy was deemed an unreliable indicator of model performance. A structured metric hierarchy was therefore established to guide both training and evaluation. During training, `val_loss` served as the sole signal for Early Stopping and learning rate reduction via `ReduceLROnPlateau`, as it provides a smooth, continuous optimization target, unlike discrete classification metrics, which produced unstable callback behaviour on minority classes. For model comparison, Macro F1-Score and Balanced Accuracy were adopted as the primary evaluative metrics, weighting all classes equally regardless of their prevalence. As a clinical safety check, Melanoma Recall was tracked explicitly to ensure the deadliest class was not systematically missed and Specificity is reported in the final evaluation table for completeness but was not used as a selection criterion. All models were compiled with `SparseCategoricalCrossentropy(from_logits=True)`,

chosen for its compatibility with integer-encoded labels and its numerical stability when the final dense layer outputs raw logits rather than softmax probabilities.

4.2. CUSTOM CNN MODELS

As a starting point, a lightweight CNN was built from scratch to establish a performance baseline. The architecture followed a three-block convolutional pipeline of increasing capacity (32, 64, and 128 filters), with a GlobalAveragePooling head feeding into a Dense(256) layer before the final 7-class logit output. Dropout was applied progressively (from 0.25 to 0.30 in conv blocks, 0.50 at the dense layer) as a precautionary regularization measure. Despite applying class weights to counteract the imbalance identified in Section 2, the model suffered from severe underfitting on minority classes, achieving `val_loss` of 0.8691, `val_accuracy` of 64.32%, `val_balanced_accuracy` of 31.85%, and a weighted F1 of 0.33, confirming that the shallow architecture lacked the feature extraction capacity to distinguish the visual complexity of 7 dermatoscopic categories. Training was manually interrupted at Epoch 25. This result directly motivated the strategic shift to ImageNet-pretrained transfer learning pipelines.

4.3. TRANSFER LEARNING WITH PRETRAINED MODELS

Given the limitations exposed by the baseline CNN, the project pivoted to transfer learning using three ImageNet-pretrained architectures: EfficientNetB0, DenseNet121, and MobileNetV2. All three share an identical classification head: GlobalAveragePooling2D followed by Dropout(0.4), Dense(256, ReLU), BatchNormalization, Dropout(0.3), and finally Dense(7, logits). This standardized head ensures that architectural differences between bases remain the sole variable across experiments.

4.3.1. Training Strategy

All models followed a two-phase fine-tuning approach. In Phase 1, the pretrained base was fully frozen and only the classification head was trained at a higher learning rate (1e-3), allowing rapid adaptation without disturbing ImageNet weights. In Phase 2, a subset of the top layers was unfrozen and the full model was fine-tuned at a reduced learning rate (1e-5) to preserve low-level feature representations while adapting higher-level features to dermatoscopic patterns.

4.3.2. EfficientNetB0

EfficientNetB0 emerged as the strongest performer. Its compound scaling (balancing depth, width, and resolution simultaneously) proved well-suited to extracting the subtle morphological features that distinguish dermatoscopic classes. Phase 2 unfroze the top 30 layers. Overfitting was mitigated through the combined effect of the augmented training set, progressive dropout, BatchNormalization, and EarlyStopping.

4.3.3. DenseNet121

DenseNet121 was trained with full base unfreeze from Epoch 1, diverging from the standard two-phase approach. This decision followed observed instability in the standard version, where combining the already-balanced offline augmentation with explicit class weights caused a double-penalization effect on minority classes, producing volatile gradients and training crashes. Removing class weights and unfreezing the entire base from the start resolved this instability, achieving val_loss of 0.8093, val_accuracy of 71.35%, val_balanced_accuracy of 57.23%, and a weighted F1 of 0.73.

4.3.4. MobileNetV2

MobileNetV2 was the weakest of the three. Its depthwise separable convolution design, optimized for mobile deployment, lacks the parametric depth required to capture the fine-grained textural differences between 7 dermatoscopic categories. Phase 2 unfroze the top 20 layers, achieving val_loss of 0.9634, val_accuracy of 65.54%, val_balanced_accuracy of 38.05%, and a weighted F1 of 0.37, barely surpassing the custom CNN baseline and confirming that architectural capacity was the limiting factor.

4.4. ENRICHED MODEL: EFFICIENTNETB1 + METADATA

Building on the strongest pretrained result, a multimodal architecture was developed combining EfficientNetB1 with structured clinical metadata, following a late fusion approach.

4.4.1. Architecture

The model consists of two parallel branches merged before the final classification layer. The image branch processes 224×224 dermoscopic images through the EfficientNetB1 backbone, followed by GlobalAveragePooling2D and a Dense(256, ReLU) and Dropout(0.4) block[MG1], yielding a 256-dimensional feature vector. The metadata branch takes a separate numeric input vector encoding three clinical variables: sex (one-hot), anatomical location (one-hot across 15 sites), and age (min-max normalized); through a small MLP: Dense(64), BatchNormalization, and Dense(32[MG2]). The two branches are concatenated into a 288-dimensional vector, passed through Dense(128), and fed into the final Dense(7) logit output.

Phase 1 froze the EfficientNetB1 backbone and trained only the fusion head at LR=1e-3 for 15 epochs. Phase 2 fully unfroze the entire backbone and fine-tuned at LR=1e-5 for 20 epochs, a more aggressive strategy than the standalone pretrained models.

4.4.2. Limitations

While the project successfully deployed a high-performing **EfficientNetB0** image-only model, the transition to a **Late Fusion Multimodal** system encountered significant technical and logistical barriers that prevented final execution.

4.5. FINAL COMPARISON

To compare all developed models systematically, a unified evaluation table (Table 3) was compiled covering architecture parameters, training configuration, and performance across Macro F1-Score, Balanced Accuracy, Melanoma Recall, and Macro AUC-ROC. A visual comparison was additionally produced via `plot_comparison()`, generating side-by-side bar charts for Macro AUC-ROC and Melanoma Recall across all models, with a hardcoded 0.80 threshold line marking the minimum acceptable clinical safety target for melanoma detection. EfficientNetB0 was selected as the best performing model, achieving approximately 0.65 `val_loss`, 78% `val_accuracy`, 70% balanced accuracy, and 0.81 Macro F1-Score, the strongest results across all architectures explored. Its compound scaling design proved well matched to the visual complexity of dermatoscopic classification, outperforming both DenseNet121 and MobileNetV2 on minority-class sensitive metrics while maintaining stable, non-overfitting training curves. MobileNetV2 ranked last, consistent with its architectural limitations discussed in Section 4.3. DenseNet121 occupied an intermediate position, with its modified training strategy partially recovering from early instability. The metadata-enriched EfficientNetB1 model did not demonstrably improve over the standalone image-only baseline, suggesting that the clinical variables available, age, sex, and anatomical location, did not provide sufficient discriminative signal beyond what the image backbone already captured, or that training time was insufficient for stable fusion convergence.

5. CONCLUSION

This study demonstrates that Deep Learning, specifically through Transfer Learning, provides a robust framework for dermatoscopic image analysis. While a custom CNN proved insufficient for the visual complexity of the HAM10000 dataset, leveraging EfficientNetB0 resulted in the strongest performance, achieving a Balanced Accuracy of 70% and a Macro F1-Score of 0.81.

Key to this success was a targeted preprocessing pipeline, including the DullRazor algorithm for artifact removal and a square root balancing strategy for offline augmentation, which stabilized training across imbalanced classes. Notably, the top-performing model met the clinical safety threshold with a Melanoma Recall exceeding 0.80. While the integration of clinical metadata did not significantly outperform image-only models, the project successfully established an end-to-end workflow capable of detecting subtle morphological patterns essential for dermatological diagnosis. Future work should focus on refining multimodal fusion and exploring Transformer-based architectures to further enhance diagnostic precision.

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APPENDIX

TABLES

Code	Diagnosis
nv	Melanocytic nevi
mel	Melanoma
bkl	Benign keratosis-like lesions
bcc	Basal cell carcinoma
akiec	Actinic keratoses / Bowen's disease
vasc	Vascular lesions
df	Dermatofibroma

Table 1: Class codes and diagnostic names for the 7 skin lesions.

	Offline Augmentation	Online Augmentation
Dataset size	12621	9214
Pre-generation time	1 minute	-
Average batch time	0.5709	1.0023
Avg epoch time (s)	62.7	107.5
Final val accuracy	0.6597	0.6710
Peak val accuracy	0.6936	0.6710

Table 2: Analysis of the comparison metrics between offline and online augmentation

FIGURES

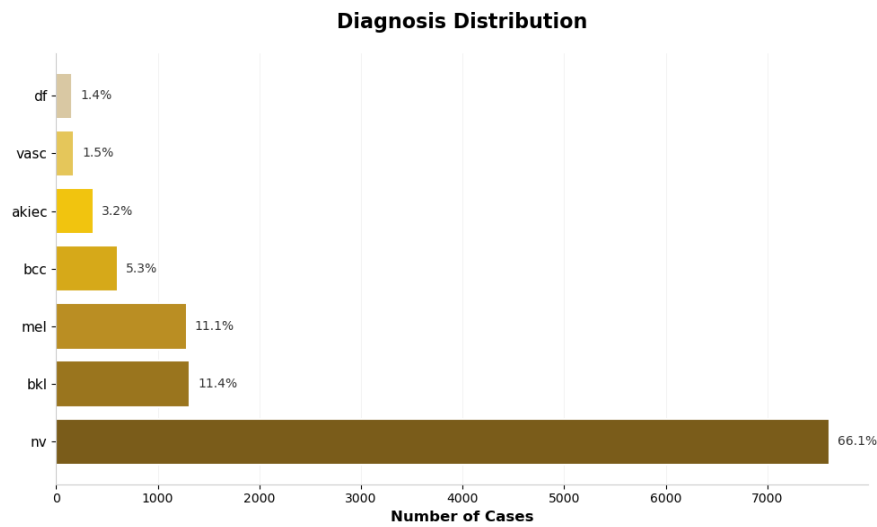


Figure 1 – Diagnosis Class Distribution

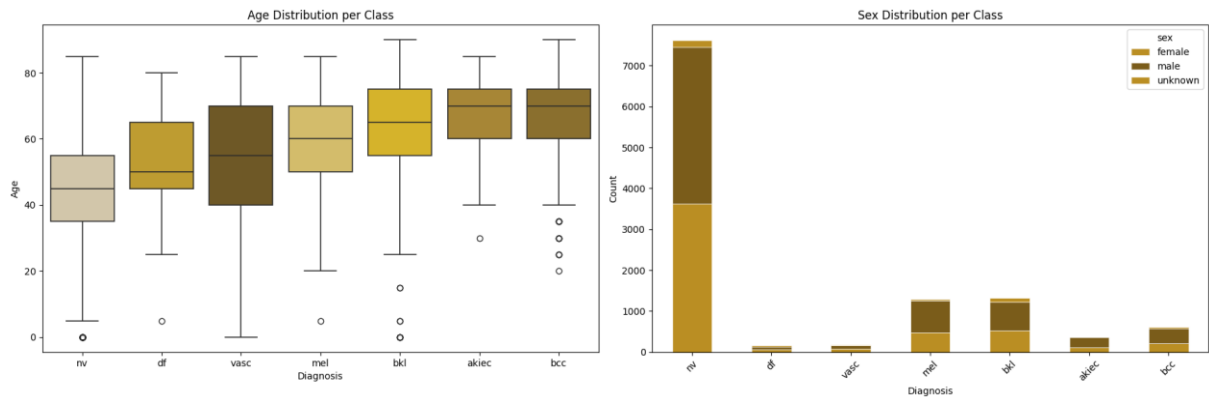


Figure 2 – Age and sex distribution per class

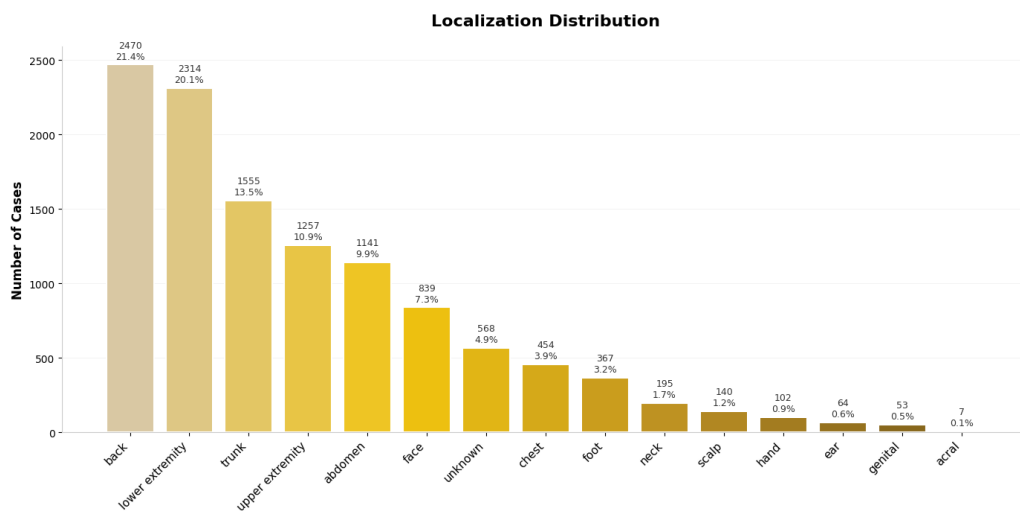


Figure 3 - Number of cases per body localization



Figure 4 – Average image per class diagnosis

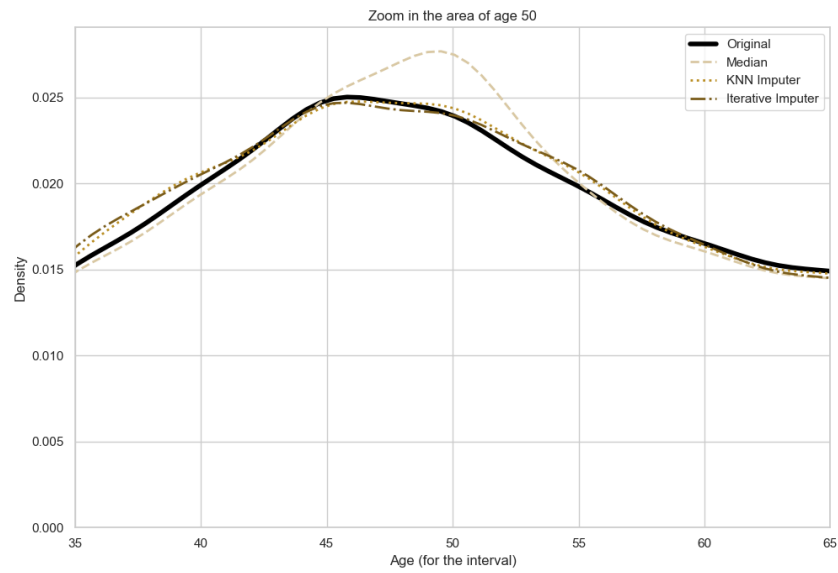


Figure 5 – Missing Values Imputation Strategies Comparison

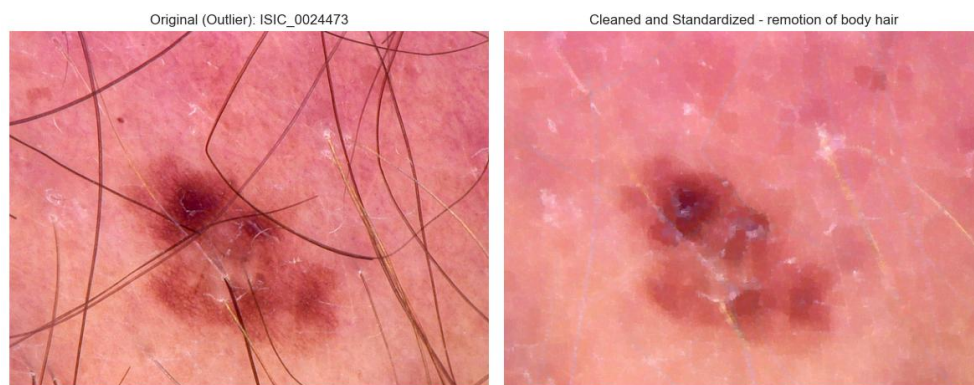


Figure 6 – Sample lesion, before and after applying the DullRazor algorithm

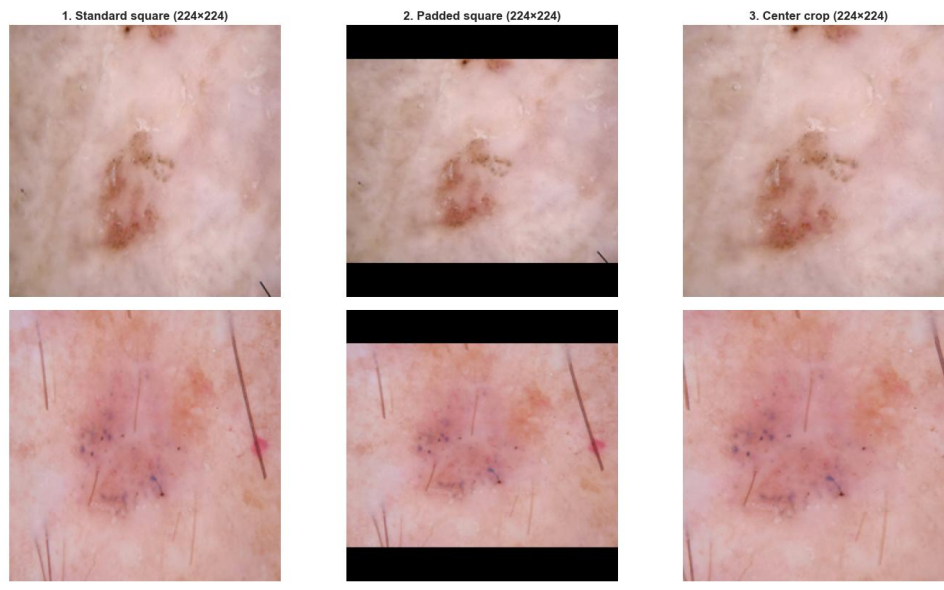


Figure 7 – Resizing Approaches Comparison

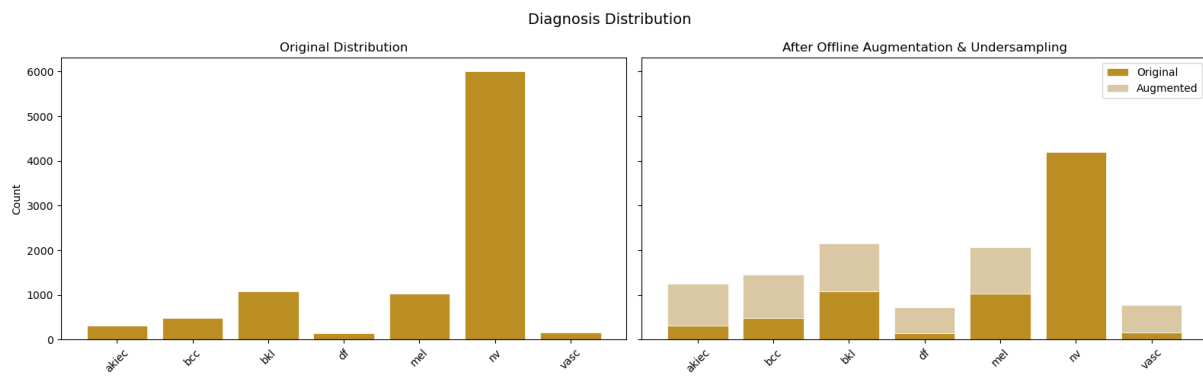


Figure 8 – Class distribution after data augmentation and undersampling